```
  private static final int NUM_HOSTS = 3;
  private static final int NUM_VMS = 12;
  private static final int NUM_CLOUDLETS = 100; // Ubah sesuai skenario (100/300/500)
  private static final long SEED = 42;
  private static final int MAX_ITERATIONS = 500;
  private static final String NEIGHBOURHOOD = "2-exchange"; // atau "insert"
    vm.setRam(2048)
      .setBw(1000)
      .setSize(50000)
      .setCloudletScheduler(new CloudletSchedulerTimeShared());
    for (int i = 0; i < n; i++) {
      long length = 500 + (long) (rand.nextDouble() * 4500);
      long fileSize = 300 + (long) (rand.nextDouble() * 2700);
      CloudletSimple cloudlet = new CloudletSimple(i, length, 1);
      cloudlet.setFileSize(fileSize)
            .setOutputSize(fileSize)
            .setUtilizationModel(utilization);
      list.add(cloudlet);
[0;39m[34mINFO  Simulation: No more future events
[0;39m[34mINFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.
[0;39m[34mINFO
================== Simulation finished at time 15,37 ==================
[0;39m
=== HASIL SIMULASI ===
Makespan         : 15,26 detik
Total Cost       : $0,2909
Avg Waiting Time : 0,10 detik
```

```
  private static final int NUM_HOSTS = 3;
  private static final int NUM_VMS = 12;
  private static final int NUM_CLOUDLETS = 300; // Ubah sesuai skenario (100/300/500)
  private static final long SEED = 42;
  private static final int MAX_ITERATIONS = 500;
  private static final String NEIGHBOURHOOD = "2-exchange"; // atau "insert"
    vm.setRam(2048)
      .setBw(1000)
      .setSize(50000)
      .setCloudletScheduler(new CloudletSchedulerTimeShared());
    for (int i = 0; i < n; i++) {
      long length = 500 + (long) (rand.nextDouble() * 4500);
      long fileSize = 300 + (long) (rand.nextDouble() * 2700);
      CloudletSimple cloudlet = new CloudletSimple(i, length, 1);
      cloudlet.setFileSize(fileSize)
            .setOutputSize(fileSize)
            .setUtilizationModel(utilization);
[0;39m[34mINFO  Simulation: No more future events
[0;39m[34mINFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.
[0;39m[34mINFO
================== Simulation finished at time 48,44 ==================
[0;39m
=== HASIL SIMULASI ===
Makespan         : 48,33 detik
Total Cost       : $0,5135
Avg Waiting Time : 0,10 detik
```

```
  private static final int NUM_HOSTS = 3;
  private static final int NUM_VMS = 12;
```

```
  private static final int NUM_CLOUDLETS = 500; // Ubah sesuai skenario (100/300/500)
  private static final long SEED = 42;
  private static final int MAX_ITERATIONS = 500;
  private static final String NEIGHBOURHOOD = "2-exchange"; // atau "insert"
    vm.setRam(2048)
      .setBw(1000)
      .setSize(50000)
      .setCloudletScheduler(new CloudletSchedulerTimeShared());
    for (int i = 0; i < n; i++) {
      long length = 500 + (long) (rand.nextDouble() * 4500);
      long fileSize = 300 + (long) (rand.nextDouble() * 2700);
      CloudletSimple cloudlet = new CloudletSimple(i, length, 1);
      cloudlet.setFileSize(fileSize)
            .setOutputSize(fileSize)
            .setUtilizationModel(utilization);
[0;39m[34mINFO  Simulation: No more future events
[0;39m[34mINFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.
[0;39m[34mINFO
================== Simulation finished at time 82,66 ==================
[0;39m
=== HASIL SIMULASI ===
Makespan        : 82,55 detik
Total Cost      : $1,2904
Avg Waiting Time : 0,10 detik
```

```
  private static final int NUM_HOSTS = 3;
  private static final int NUM_VMS = 12;
  private static final int NUM_CLOUDLETS = 100; // Ubah sesuai skenario (100/300/500)
  private static final long SEED = 42;
  private static final int MAX_ITERATIONS = 500;
  private static final String NEIGHBOURHOOD = "insert"; // atau "insert" "2-exchange"
    vm.setRam(2048)
      .setBw(1000)
      .setSize(50000)
      .setCloudletScheduler(new CloudletSchedulerTimeShared());
    for (int i = 0; i < n; i++) {
      long length = 500 + (long) (rand.nextDouble() * 4500);
      long fileSize = 300 + (long) (rand.nextDouble() * 2700);
      CloudletSimple cloudlet = new CloudletSimple(i, length, 1);
      cloudlet.setFileSize(fileSize)
            .setOutputSize(fileSize)
            .setUtilizationModel(utilization);
[0;39m[34mINFO  Simulation: No more future events
[0;39m[34mINFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.
[0;39m[34mINFO
================== Simulation finished at time 15,47 ==================
[0;39m
=== HASIL SIMULASI ===
Makespan        : 15,36 detik
Total Cost      : $0,2549
Avg Waiting Time : 0,10 detik
```

```
  private static final int NUM_HOSTS = 3;
  private static final int NUM_VMS = 12;
  private static final int NUM_CLOUDLETS = 300; // Ubah sesuai skenario (100/300/500)
  private static final long SEED = 42;
  private static final int MAX_ITERATIONS = 500;
  private static final String NEIGHBOURHOOD = "insert"; // atau "insert" "2-exchange"
    vm.setRam(2048)
```

```
        .setBw(1000)
        .setSize(50000)
        .setCloudletScheduler(new CloudletSchedulerTimeShared());
    for (int i = 0; i < n; i++) {
        long length = 500 + (long) (rand.nextDouble() * 4500);
        long fileSize = 300 + (long) (rand.nextDouble() * 2700);
        CloudletSimple cloudlet = new CloudletSimple(i, length, 1);
        cloudlet.setFileSize(fileSize)
                .setOutputSize(fileSize)
                .setUtilizationModel(utilization);
[0;39m[34mINFO  Simulation: No more future events
[0;39m[34mINFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.
[0;39m[34mINFO
================== Simulation finished at time 45,11 ==================
[0;39m
=== HASIL SIMULASI ===
Makespan        : 45,00 detik
Total Cost      : $-0,1640
Avg Waiting Time : 0,10 detik
```

```
  private static final int NUM_HOSTS = 3;
  private static final int NUM_VMS = 12;
  private static final int NUM_CLOUDLETS = 500; // Ubah sesuai skenario (100/300/500)
  private static final long SEED = 42;
  private static final int MAX_ITERATIONS = 500;
  private static final String NEIGHBOURHOOD = "insert"; // atau "insert" "2-exchange"
    vm.setRam(2048)
      .setBw(1000)
      .setSize(50000)
      .setCloudletScheduler(new CloudletSchedulerTimeShared());
    for (int i = 0; i < n; i++) {
        long length = 500 + (long) (rand.nextDouble() * 4500);
        long fileSize = 300 + (long) (rand.nextDouble() * 2700);
        CloudletSimple cloudlet = new CloudletSimple(i, length, 1);
        cloudlet.setFileSize(fileSize)
                .setOutputSize(fileSize)
                .setUtilizationModel(utilization);
[0;39m[34mINFO  Simulation: No more future events
[0;39m[34mINFO  CloudInformationService0: Notify all CloudSim Plus entities to shutdown.
[0;39m[34mINFO
================== Simulation finished at time 82,08 ==================
[0;39m
=== HASIL SIMULASI ===
Makespan        : 81,97 detik
Total Cost      : $0,5418
Avg Waiting Time : 0,10 detik
```